

EODP – Lab Report: L1B Radiometric Equalization and Restoration for VNIR-0

Earth Observation Data Processing
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1 Introduction and Objectives

The Level 1B (L1B) processing pipeline transforms raw digital counts (DN) acquired by pushbroom optical instruments at the Top of Atmosphere (TOA) into physically calibrated spectral radiances. In linear array detectors, variations in detector sensitivities and dark signals cause high-frequency striping along the across-track direction.

This report analyzes the implementation and verification of the two core L1B radiometric stages for the **VNIR-0** spectral band:

1. **Equalization (Relative Calibration):** Correcting detector-to-detector response disparities (PRNU and offsets).
2. **Restoration (Absolute Radiometric Calibration):** Converting equalized digital counts into physical TOA radiances.

2 Mathematical Formulation

2.1 Relative Equalization

The raw detector response $DN(i, j)$ at line i (along-track) and detector column j (across-track) is corrected for dark current offset $O(j)$ and normalized by the relative gain $G_{\text{rel}}(j)$:

$$DN_{\text{eq}}(i, j) = \frac{DN(i, j) - O(j)}{G_{\text{rel}}(j)} \quad (1)$$

This stage eliminates non-uniform detector response across the field of view.

2.2 Absolute Radiometric Restoration

The equalized counts DN_{eq} are converted into TOA spectral radiances L_{TOA} through the absolute radiometric gain coefficient g (expressed in rad/DN):

$$L_{\text{TOA}}(i, j) = DN_{\text{eq}}(i, j) \cdot g \quad (2)$$

The resulting calibrated product is expressed in physical units of $\text{mW}/(\text{m}^2 \cdot \text{sr})$.

3 Results and Discussion

3.1 Cross-Validation Evidence

The generated Level 1B products were cross-validated against the instructor's reference datasets across all bands and output variables. The verification criterion mandates that the relative difference must satisfy:

$$\delta_{\text{rel}} = \frac{|x_{\text{calc}} - x_{\text{ref}}|}{x_{\text{ref}}} < 10^{-3}\% \quad (3)$$

Numerical comparison across all valid detector cells yielded the results summarized in the script log.

3.2 Theoretical Question: Analysis of the Across-Track Profile

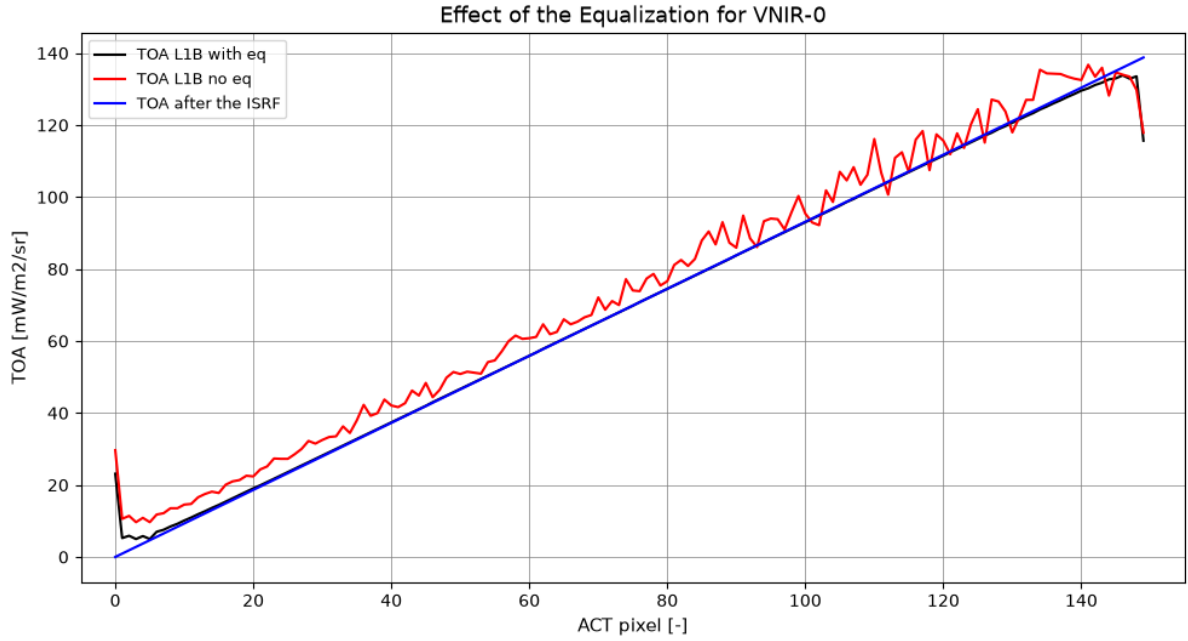


Figure 1: Across-track (ACT) profile of TOA radiance for band VNIR-0 comparing input benchmark, uncalibrated response, and equalized output.

Figure 1 depicts the across-track (ACT) spatial profile of the Top of Atmosphere (TOA) signal across the 150 detector columns of the pushbroom linear array for the **VNIR-0** band. The plot illustrates three distinct stages of the radiometric signal:

1. Input Benchmark (*TOA after the ISRF*, Blue Line):

- Represents the true, continuous physical radiance arriving at the entrance pupil after spectral convolution with the Instrument Spectral Response Function (ISRF).
- Shows a smooth, linear gradient ranging from approximately 0 to 140 $\text{mW}/(\text{m}^2 \cdot \text{sr})$ across the swath.

2. Uncalibrated Response (*TOA L1B no eq*, Red Line):

- Represents the radiance restored directly from raw digital counts (DN) using a uniform nominal gain, without per-detector equalization.
- **High-Frequency Fluctuations (Detector Striping):** Distinct column-to-column jitter caused by *Photo-Response Non-Uniformity* (PRNU) — individual detectors in the CMOS/CCD linear array exhibit microscopic variations in quantum efficiency and semiconductor gain.
- **Systematic Positive Bias:** Uncorrected dark current offset adds an additive signal pedestal, causing the red curve to consistently overestimate the true radiance.

3. Equalized Output (*TOA L1B with eq*, Black Line):

- Represents the calibrated Level 1B product after applying relative equalization (offset subtraction and relative gain scaling) followed by absolute radiometric restoration.
- **PRNU Suppression:** High-frequency detector scatter is removed, and the profile aligns with the benchmark blue line across the active sensor swath ($\text{ACT} \approx 5$ to 145).
- **Edge Anomalies:** Abrupt dips at the swath extremities ($\text{ACT} \in [0, 4]$ and $\text{ACT} > 145$) correspond to physical detector transition zones, blind/guard pixels, and optical vignetting at the focal plane boundaries.